For the **"Shallow Parsing and Tools for NLP"** module, the lab experiments should directly map to the theory topics and progressively build NLP skills using Python, NLTK, spaCy, and WordNet.

NLP Laboratory Experiments

Shallow Parsing and Tools for NLP

Experiment 1: Tokenization of Text Data

Aim: Develop a Python program to perform sentence and word tokenization on textual data.

Concepts Covered: Tokenization

Tools: NLTK

Experiment 2: Stop Word Removal and Text Preprocessing

Aim: Implement basic text preprocessing techniques including stop-word removal and punctuation filtering.

Concepts Covered: NLP Preprocessing

Tools: NLTK

Experiment 3: Part-of-Speech (POS) Tagging

Aim: Develop a Python program to identify and classify words into their grammatical categories.

Concepts Covered: Tokenization & POS Tagging

Tools: NLTK, spaCy

Experiment 4: Stemming using Porter Stemmer

Aim: Implement stemming algorithms to obtain the root form of words.

Concepts Covered: Stemming

Tools: NLTK Porter Stemmer

Experiment 5: Lemmatization using WordNet

Aim: Develop a Python program to perform lemmatization and compare the results with stemming.

Concepts Covered: Lemmatizer & Stemming

Tools: NLTK WordNet Lemmatizer

Experiment 6: Morphological Analysis of Words

Aim: Develop a Python program to perform morphological analysis by identifying prefixes, suffixes, and root words.

Concepts Covered: Morphological Analysis

Tools: NLTK, Custom Rules

Experiment 7: Chunking for Shallow Parsing

Aim: Implement chunking to identify noun phrases, verb phrases, and prepositional phrases from sentences.

Concepts Covered: Chunking

Tools: NLTK RegexpParser

Experiment 8: Multi-Word Expression (MWE) Identification

Aim: Develop a Python program to detect and extract multi-word expressions from text data.

Concepts Covered: Multi-Word Expressions

Tools: NLTK MWETokenizer

Experiment 9: Named Entity Recognition (NER)

Aim: Extract named entities such as persons, organizations, locations, dates, and monetary values from textual documents.

Concepts Covered: Named Entity Recognition

Tools: spaCy, NLTK

Experiment 10: Morphological Synthesis

Aim: Develop a system to generate inflected word forms from root words based on grammatical features.

Concepts Covered: Morphological Synthesis

Tools: Rule-Based Python Implementation

Experiment 11: Word Sense Disambiguation (WSD)

Aim: Develop a Python program to determine the correct meaning of an ambiguous word based on context.

Concepts Covered: Word Sense Disambiguation

Tools: NLTK Lesk Algorithm, WordNet

Experiment 12: Universal Networking Language (UNL) Representation

Aim: Represent natural language sentences using Universal Networking Language semantic relations.

Concepts Covered: UNL

Tools: Python, Rule-Based Mapping

Dataset-Based Assignment (Disneyland Reviews)

Assignment: Apply NLP preprocessing and shallow parsing techniques on the **Disneyland Reviews Dataset**.

Dataset: https://www.kaggle.com/arushchillar/disneyland-reviews

Tasks

1. Perform Tokenization.
2. Remove Stop Words.
3. Apply Stemming.
4. Apply Lemmatization.
5. Perform POS Tagging.
6. Extract Noun Phrases using Chunking.
7. Identify Named Entities.

8. Extract Multi-Word Expressions.
9. Perform Word Sense Disambiguation on selected review terms.
10. Generate a summary report comparing the outputs.

Recommended University NLP Lab Sequence

1. Tokenization
2. Stop Word Removal
3. POS Tagging
4. Stemming
5. Lemmatization
6. Morphological Analysis
7. Chunking
8. Multi-Word Expression Extraction
9. Named Entity Recognition
10. Morphological Synthesis
11. Word Sense Disambiguation
12. Universal Networking Language Representation

This sequence aligns exactly with the syllabus topics shown in the uploaded image and can be completed as a **12-experiment NLP laboratory course**.